CONSERVATIVE CSP ENGINE: MATHEMATICAL & ORCHESTRATION PROTOCOL

The institutional deployment of Cash-Secured Puts (CSPs) requires a rigid separation of concerns: mathematical screening must be strictly deterministic, while portfolio sizing and risk management require dynamic, state-aware correlation analysis. This whitepaper details the end-to-end architecture of a modern, two-stage quantitative CSP framework. Stage 1 executes continuous Black-Scholes screening and Put-Wall gamma shielding. Stage 2 routes surviving alpha through a Multi-Agent LangGraph state machine, automating collateral allocation and generating generative-AI executive synthesis.

1. THE QUANTITATIVE FOUNDATION (STAGE 1)

The core engine evaluates a strict universe of non-leveraged mega-cap equities and liquid broad-market ETFs (e.g., ORCL, NVDA, CRWD, QQQ, SPY). Leveraged 2x/3x exchange-traded products subject to daily-reset swap decay are structurally excluded from this tier. Every candidate put contract is priced using exact, continuous Black-Scholes Greeks, anchored against live 13-Week U.S. Treasury rates (ticker: ^IRX) and individual strike-level Implied Volatility (σ). The foundational mathematical definitions governing the engine are as follows:

Inclusive Trading Horizon (DTE & Years T)

$$DTE = \max(1, (T_{exp} - T_{today}).days + 1) \qquad T = DTE / 365.0$$

Executable Bid Yield & Annualization

$$Y_{raw} = (Bid / K) \times 100 \qquad Y_{ann} = Y_{raw} \times (365.0 / DTE)$$

Black-Scholes Put Delta (Δ)

$$d_1 = [\ln(S / K) + (r + 0.5\sigma^2)T] / (\sigma\sqrt{T}) \qquad \Delta_{put} = N(d_1) - 1$$

True Breakeven & Downside Buffer %

$$BE = K - Bid \qquad Buffer_{\%} = [(S - BE) / S] \times 100$$

Before any portfolio optimization occurs, candidate contracts are evaluated across the options chain for the upcoming four Fridays. Alpha is quantified via a proprietary balanced value score: annualized yield (which decays with DTE) multiplied by total downside buffer (which scales with DTE), constrained by the risk-neutral probability of profit derived from the Black-Scholes delta.

THE 8-GATE "SUDDEN-DEATH" GAUNTLET

Every candidate option contract must pass all eight sequential boolean gates below. Failure at any single gate triggers immediate mathematical disqualification. This protocol prevents manual override biases and enforces institutional-grade capital preservation.

Gate ID & Name	Exact Code Condition	Structural Engineering Rationale
1: Horizon & Friday Rule	exp.weekday() == 4 & 1 ≤ DTE ≤ 28	Restricts entries strictly to standard Friday expirations inside the 1-to-28 day theta acceleration window; rejects mid-week chains.
2: Earnings Firewall	Days_to_Earnings > (DTE + 7) (Exempts ETF Whitelist)	Eliminates binary gap risk. Requires a full 7-calendar-day post-expiration cushion before any scheduled corporate earnings announcement.
3: Put-Wall Gamma Shield	Strike < Wall_Stk (Where Wall_Stk is Max OI within 10% OTM)	Locates the largest institutional Open Interest strike within 10% below spot and forces entries strictly underneath dealer gamma support.
4: Liquidity & Volume Floor	OI ≥ 500 & (Vol ≥ 100 if DTE ≤ 5 else Vol ≥ 20)	Ensures deep institutional book depth and active daily turnover so limit orders fill cleanly without illiquid slippage traps.
5: Quote & Slippage Filter	Bid > 0 & Ask ≥ Bid & (Ask − Bid) / Ask ≤ 0.10	Rejects corrupted zero-ask data feeds and enforces a strict 10.0% Bid-Ask spread cap to minimize entry slippage.
6: BS Delta Band	$-0.20 \le \Delta_{put} \le -0.10$	Enforces an 80% to 90% risk-neutral probability of expiring out-of-the-money, aggressively filtering high-gamma near-the-money risk.
7: T-Bill Yield Floor	Buffer ≥ Min_Buf & $Y_{ann} > r_{\hat{}IRX}$	Guarantees structural cushion and verifies the annualized premium comfortably exceeds the risk-free U.S. Treasury yield.
8: Monotonic Cross-Week	$K_{week(i)} \le K_{week(i-1)}$	Selects only the highest-scoring strike per Friday, mandating that longer-duration setups never assume higher strike risk than shorter durations.

2. MULTI-AGENT ORCHESTRATION (STAGE 2)

While the Stage 1 quantitative gauntlet successfully identifies isolated market alpha, deploying capital across multiple contracts introduces correlated portfolio risk. Evaluating a surviving list of candidate tickets in a vacuum risks the "Split-Brain Trap"—allocating multiple contracts to heavily correlated semiconductor equities simply because they all individually passed Black-Scholes delta limits.

To safely bridge the gap between raw signal generation and live execution, Stage 2 utilizes a deterministic Multi-Agent State Graph. Constructed via LangGraph architecture, this sequential pipeline operates on a shared state dictionary containing the target capital pool, current ledger exposures, and active macroeconomic calendar variables. The routing framework strictly separates validation logic into autonomous processing nodes.

NODE 1: DATA SANITY & MACRO AUDITOR

Acting as the first bridge gate, this node cross-references each mathematically qualified candidate against non-quantitative, real-world risks. It traps unverified or missing equity earnings calendars (failing closed to prevent blind entries). Furthermore, it scans the proposed duration against a forward-looking macroeconomic calendar, flagging or hard-blocking candidates if critical binary data releases (e.g., FOMC Rate Decisions, Nonfarm Payrolls, CPI) intersect with near-term expirations lacking sufficient put-wall buffer.

NODE 2: PORTFOLIO RISK & CORRELATION MANAGER

This node is responsible for global portfolio beta limits and deduplication. It maps every underlying ticker to a specific Sector Cluster (`AI_SEMI` , `MEGA_TECH_CLOUD` , `BROAD_ETF`). It strictly prohibits the simultaneous funding of multiple expirations for the same underlying ticker, effectively forcing the graph to select the single most optimal runway duration. Additionally, it enforces cluster density caps (e.g., restricting deployment to a maximum of two unique equities per high-beta sector per execution cycle).

CAPITAL SIZING & GENERATIVE SYNTHESIS

Following strict mathematical screening and correlation auditing, the surviving signals must be translated into discrete contract counts. This requires dynamic logic capable of adapting to varying account geometries without violating fractional risk mandates.

NODE 3: DYNAMIC TREASURY ALLOCATOR

Node 3 receives the correlated approvals and translates them into executable lot sizes based on highly modular, parametric risk configurations. Rather than utilizing hardcoded dollar amounts, the engine defines sizing constraints programmatically. These include:

- **Max Trade Density:** A standardized percentage ceiling (e.g., 15% of the total capital pool) limits the sizing of any single equity position.
- **Single-Lot Override:** A secondary, expanded percentage threshold (e.g., 28%) specifically engineered to allow single-contract execution on high-notional index vehicles (like QQQ or SPY) that would otherwise mathematically fail standard limits.
- **Cluster Saturation:** Hard capital ceilings applied to specific sector clusters (e.g., restricting cumulative semiconductor exposure to 30% of total pool capital).

Calculations strictly floor to integers; partial contract geometries are discarded. Trades exceeding parameters output as *NO_GO*.

NODE 4: LLM EXECUTIVE CHAIR SYNTHESIS

The final layer introduces large language model (LLM) intelligence, executed strictly in a readout capacity to prevent trading hallucinations. Utilizing a generative AI endpoint (such as the Gemini Pro or Flash models), Node 4 ingests the deterministic math, sizing decisions, and audit flags from Nodes 1-3. Guided by strict system instructions, the LLM produces a concise, institutional-grade qualitative memo summarizing the deployed annualized yield, defensive Put-Wall positioning, and specific capital preservation rationales for rejected candidates.

3. PRODUCTION INFRASTRUCTURE & DEPLOYMENT

Theoretical screening math holds no value without resilient production architecture. To ensure data persistence, idempotent execution, and automated lifecycle tracking, the entire Python codebase abandons fragile flat files in favor of built-in, ACID-compliant relational databases and secure local web serving.

The system anchors state using a local SQLite Vault containing four strictly defined tables:

- **scan_runs:** Logs macroscopic execution metadata, including risk-free rates, deployed capital percentages, dry powder reserves, and the final LLM Executive Chair commentary for every discrete run.
- **stage1_candidates:** Archives the complete, unfiltered quantitative leaderboard prior to multi-agent sizing.
- **committee_tickets:** Records the final GO/NO_GO verdicts for every screened candidate, documenting explicit contract quantities, collateral locks, limit prices, and GTC exit targets.
- **official_ledger:** The ultimate source of truth for active portfolio positions. It features automated chronological grading logic that queries historical market closing data to classify expired trades as out-of-the-money wins or intrinsic losses, cleanly updating cumulative P&L metrics.

Execution is managed via a lightweight, dependency-free Local Python Command Center. This interface provides an interactive, dark-mode graphical overlay to the underlying SQLite tables, allowing the operator to cycle through historical multi-agent decisions, trigger live asynchronous option scans, and manually log executions directly to the ledger without cloud-hosted bloat or external API exposure.